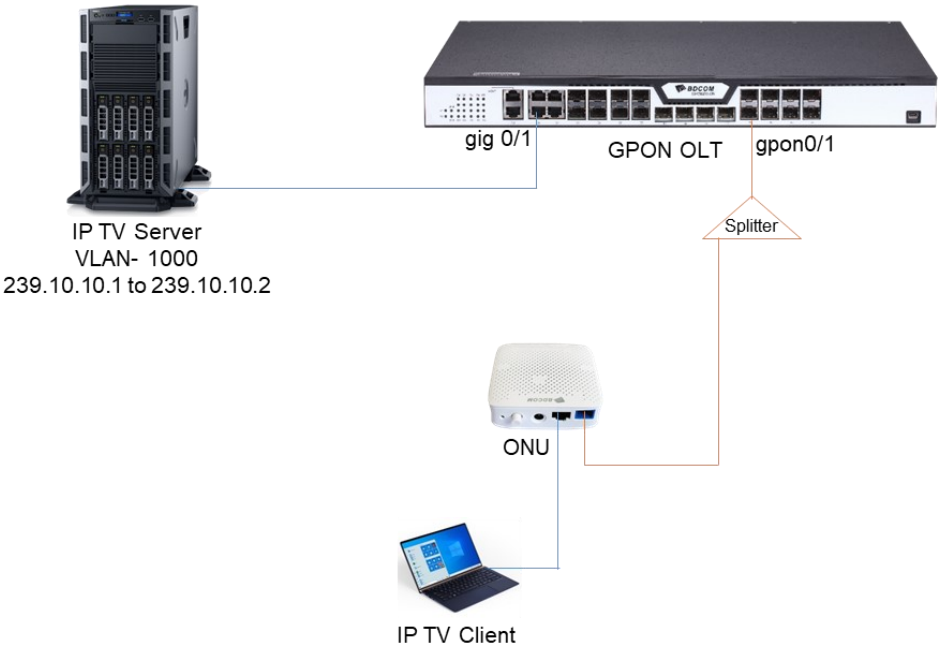
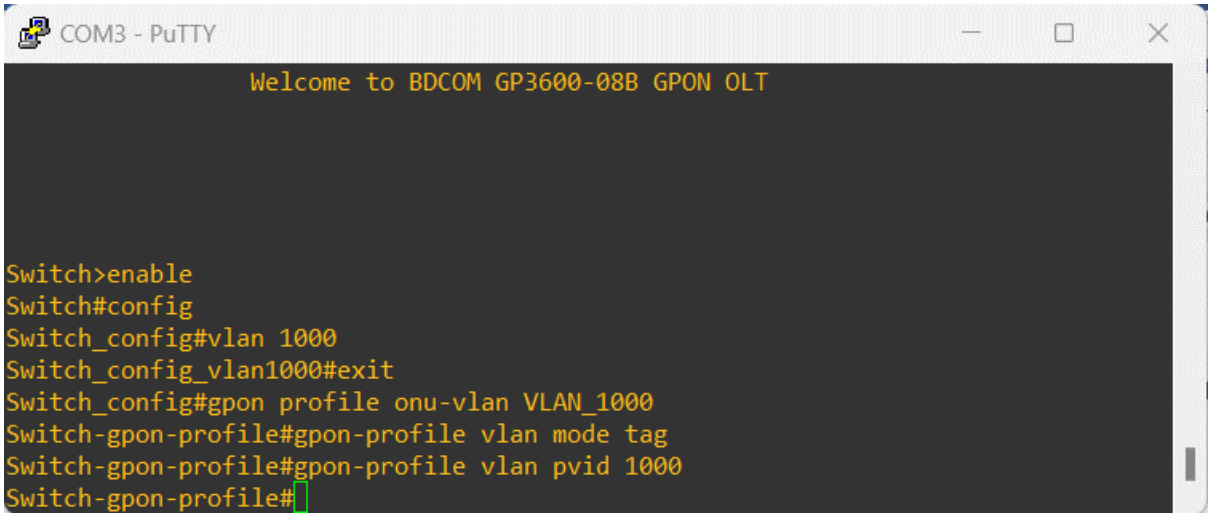
Configuring Multicast in GPON and Testing



To create a VLAN and VLAN profile in OLT:

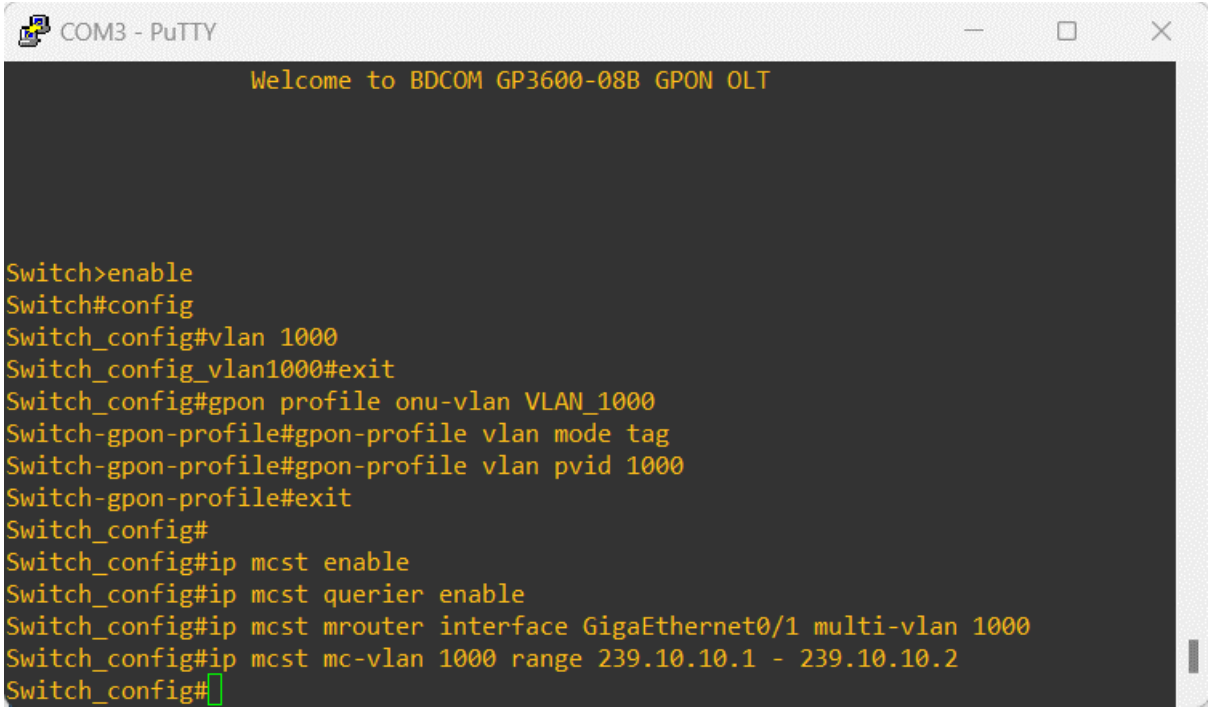
- Switch#config
- Switch_config#vlan 1000
- Switch_config_vlan1000#exit
- Switch_config#gpon profile onu-vlan **VLAN_1000**
- Switch-gpon-profile#gpon-profile vlan mode tag
- Switch-gpon-profile#gpon-profile vlan pvid 1000



Note: Here the Multicast VLAN is 1000. We may create any VLAN if we want. The VLAN profile name is **VLAN_1000**

To enable Multicast in OLT:

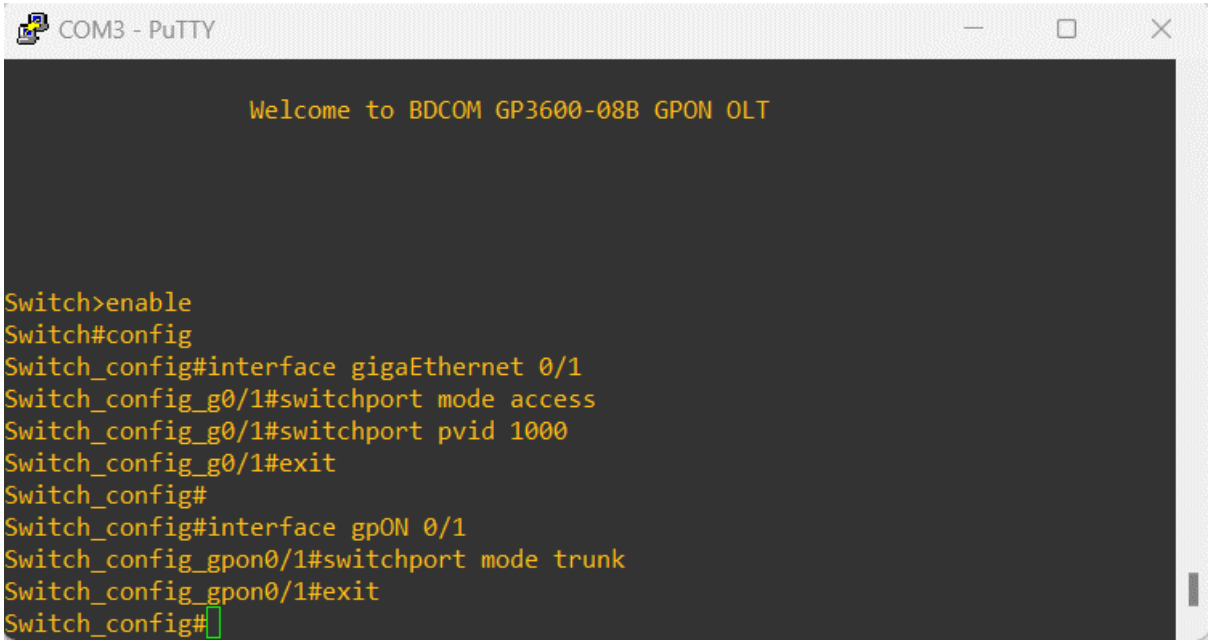
- Switch#config
- Switch_config#ip mcst enable
- Switch_config#ip mcst querier enable
- Switch_config#ip mcst mrouter interface GigaEthernet0/1 multi-vlan 1000
- Switch_config#ip mcst mc-vlan 1000 range 239.10.10.1 - 239.10.10.2



Note: Here we consider our uplink port is Giga0/1 and the IP range is 239.10.10.1 - 239.10.10.2. We can change it if we you want.

To configure the interface in OLT:

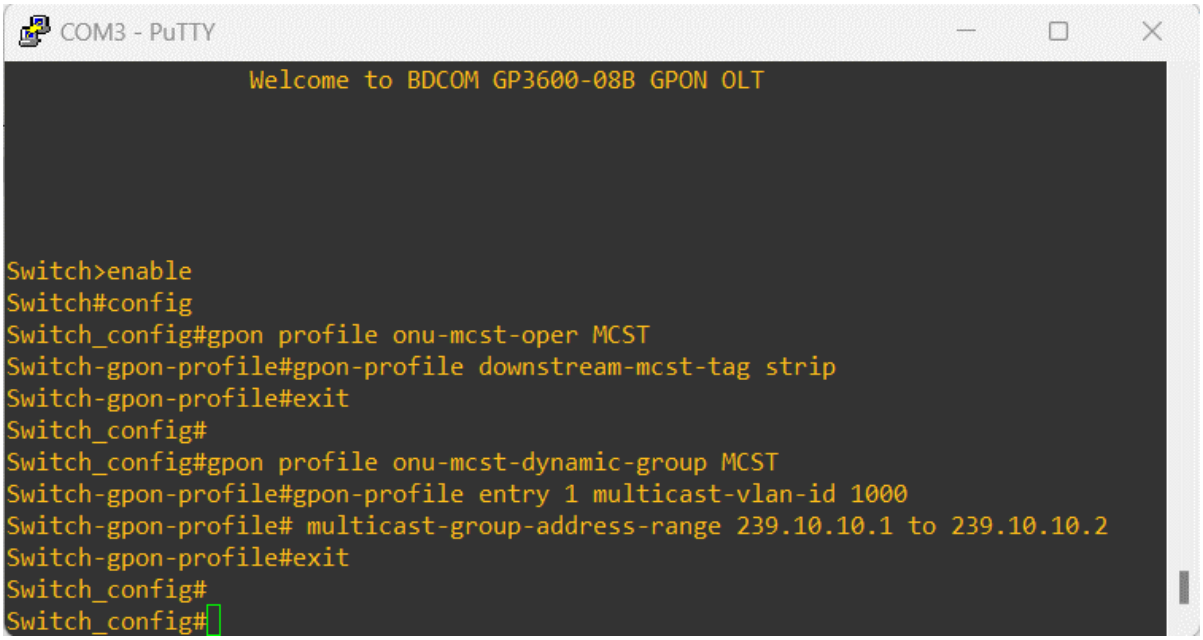
- Switch#config
- Switch_config#interface gigaEthernet 0/1
- Switch_config_g0/1#switchport mode access
- Switch_config_g0/1#switchport pvid 1000
- Switch_config_g0/1#exit
- Switch_config#interface gpoN 0/1
- Switch_config_gpon0/1#switchport mode trunk
- Switch_config_gpon0/1#exit
- Switch_config#write all



Note: Here we consider our uplink port is Giga0/1 and PON port is gpon0/1

To create a Multicast profile in OLT:

- Switch#config
- Switch_config#gpon profile onu-mcst-oper **MCST**
- Switch-gpon-profile#gpon-profile downstream-mcst-tag strip
- Switch-gpon-profile#exit
- Switch_config#gpon profile onu-mcst-dynamic-group **MCST**
- Switch-gpon-profile#gpon-profile entry 1 multicast-vlan-id 1000
- Switch-gpon-profile#gpon-profile entry 1 multicast-group-address-range 239.10.10.1 to 239.10.10.2



Note: Here the Multicast profile name is **MCST** and we consider the IP range is 239.10.10.1 to 239.10.10.2

To apply the profile on a specific ONU:

- Switch#config
- Switch_config#interface gpON 0/1:1
- Switch_config_gpon0/1:1#gpon onu uni 1 vlan-profile **VLAN_1000**
- Switch_config_gpon0/1:1#gpon onu uni 1 mcst-oper-profile **MCST**
- Switch_config_gpon0/1:1#gpon onu uni 1 mcst-dynamic-group-profile **MCST**
- Switch_config_gpon0/1:1#exit
- Switch_config#write all

```
COM3 - PuTTY

                Welcome to BDCOM GP3600-08B GPON OLT




Switch>Jan  1 00:16:40 User admin logged in from  on console 0
enable
Switch#Jan  1 00:16:41 User admin enter privilege mode from console 0, level = 15

Switch#config
Switch_config#interface gpON 0/1:1
Switch_config_gpon0/1:1#gpon onu uni 1 vlan-profile VLAN_1000
Switch_config_gpon0/1:1#gpon onu uni 1 mcst-oper-profile MCST
Switch_config_gpon0/1:1#gpon onu uni 1 mcst-dynamic-group-profile MCST
Switch_config_gpon0/1:1#exit
Switch_config#
```

To check the configuration run the command ***show running-config interface gpON 0/1:1***

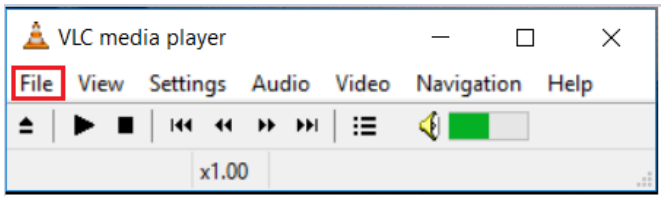
```
COM3 - PuTTY
Switch_config#show running-config interface gpON 0/1:1
Building configuration...

Current configuration:
!
interface GPON0/1:1
 gpon onu model-id GP1702-1Gv2
 gpon onu tcont-virtual-port-bind-profile tvbind-default
 gpon onu flow-mapping-profile flow-mapping-default
 gpon onu virtual-port 1 gem-port 257
 gpon onu tcont 1 alloc-id 257
 gpon onu uni 1 vlan-profile VLAN_1000
 gpon onu uni 1 mcst-oper-profile MCST
 gpon onu uni 1 mcst-dynamic-group-profile MCST
Switch_config#
```

Note: In this configuration we have applied the VLAN profile **VLAN_1000** and Multicast profile **MCST** on the specific ONU of gpon0/1:1

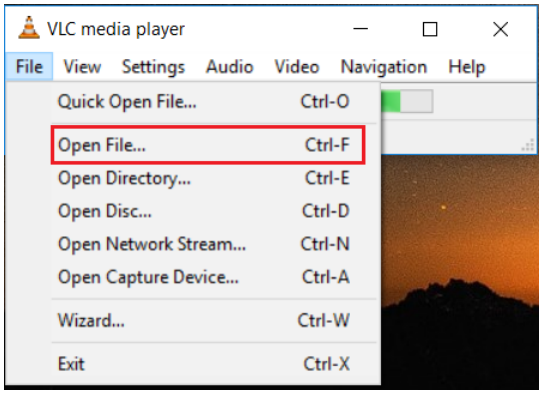
Multicast testing procedure:

To configure client side:

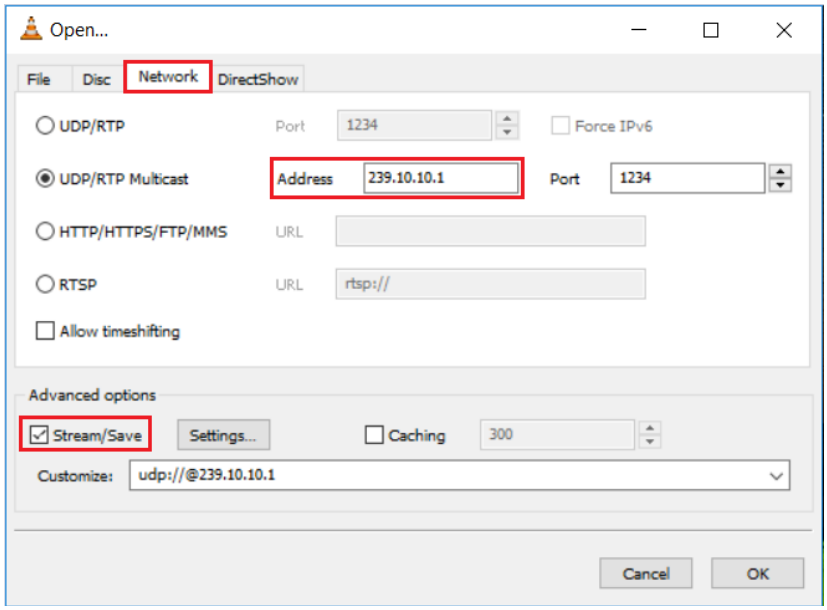
1. Install VLC media player to your client PC
2. Open VLC media player and click on **File**



3. Click on **Open File**



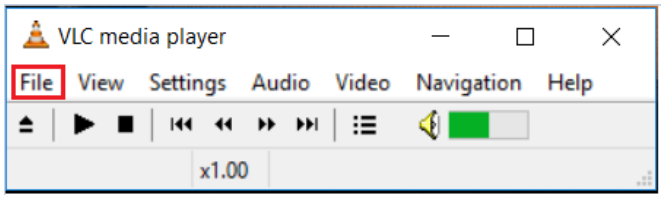
4. Click on **Network**
5. Select UDP/RTP Multicast and configure your server IP Address. Here, we can consider the server IP is 239.10.10.1
6. **Stream/Save** should be tick marked



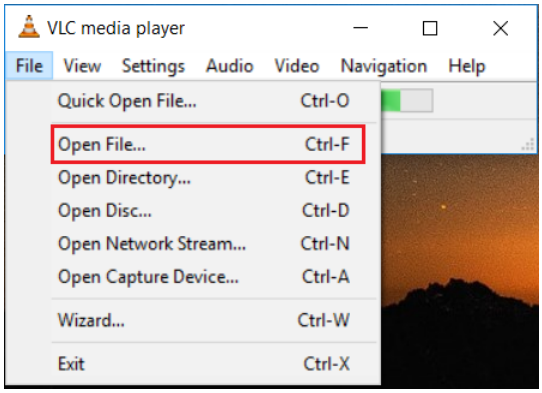
Note to student: Here we used VLC media player version 0.8.5

To configure the Server:

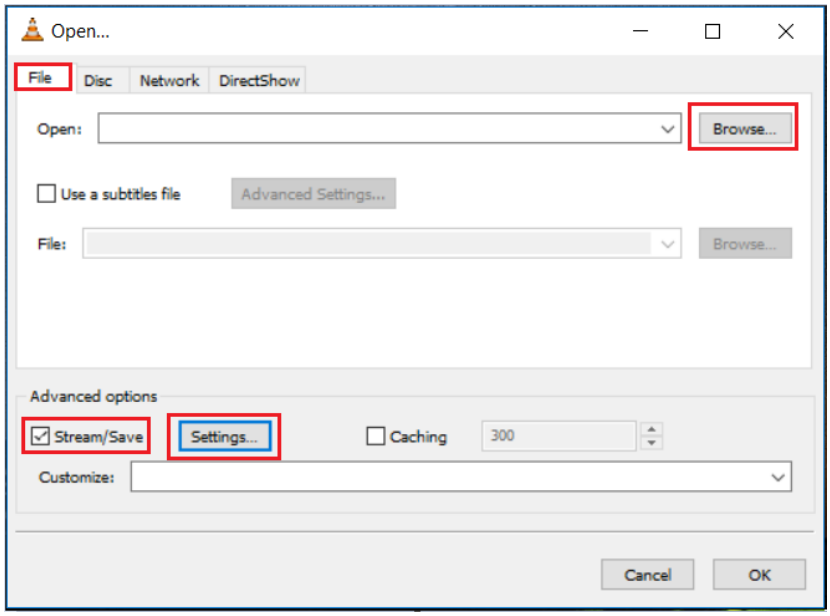
1. Install VLC media player to your client PC
2. Open VLC media player and click on **File**



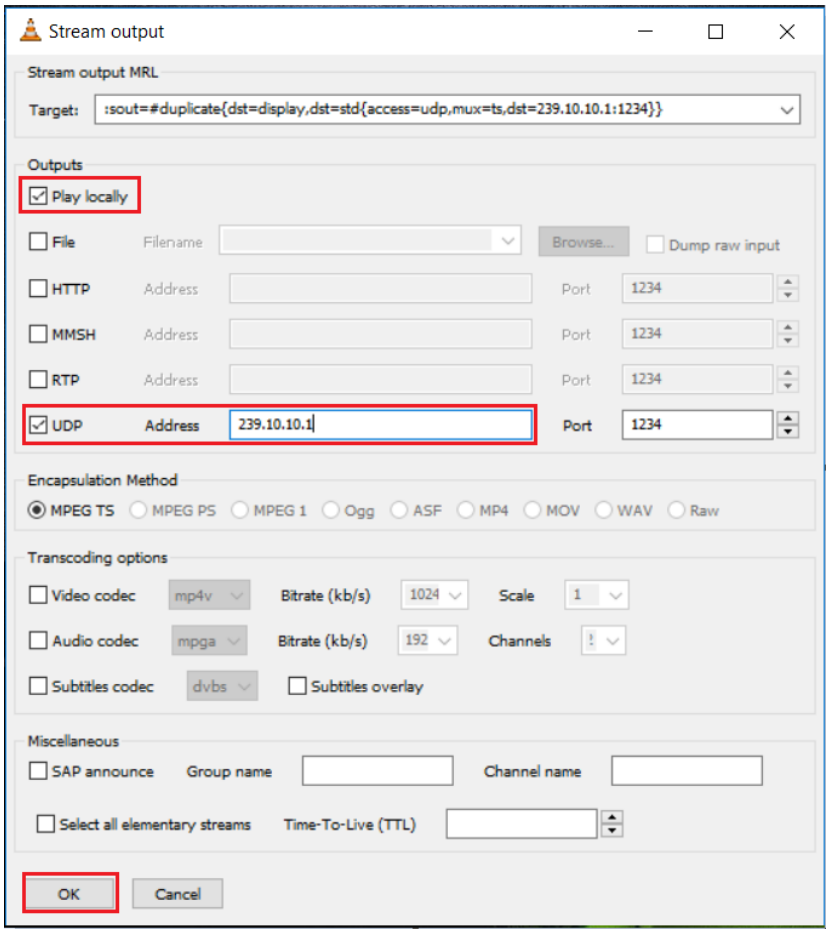
3. Click on **Open File**



4. Select **File** tab
5. Browse the video file that you want to show to your client
6. **Stream/Save** should be tick marked
7. Click on **Settings**

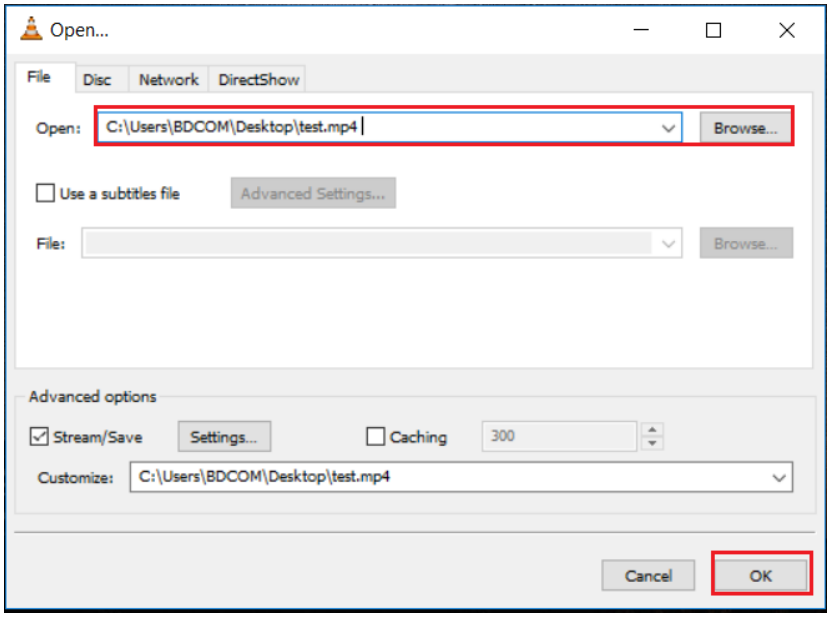


8. **Play locally** should be tick marked
9. Select **UDP** and in Address filed configure your server address
10. Click on **OK**



11. Select the video file that we would like to watch to the client side by browsing the file location

12. Click on **OK**



Note: Whenever we have done the configuration on server side the video will be plaid at the client side.

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